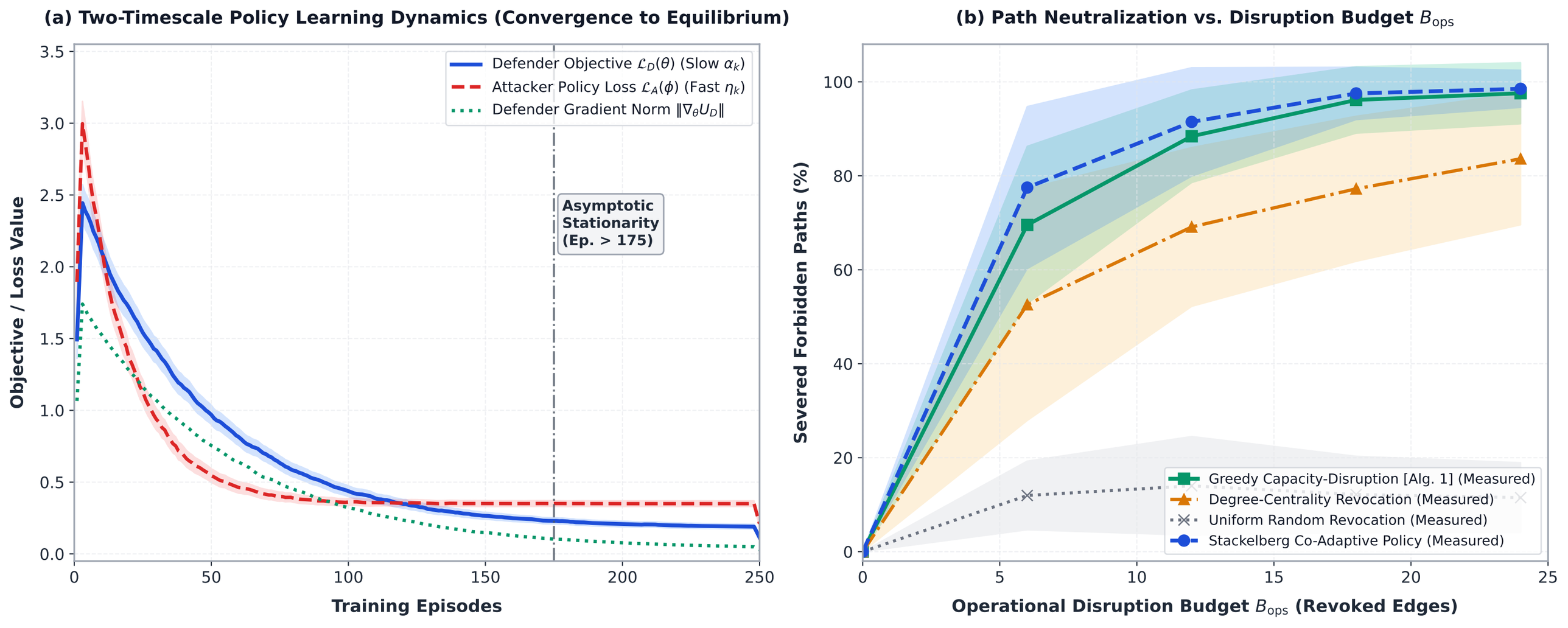

(a) Two-Timescale Policy Learning Dynamics (Convergence to Equilibrium)
Defender Objective $\mathcal{L}_D(\theta)$ (Slow α_k)
Attacker Policy Loss $\mathcal{L}_A(\phi)$ (Fast η_k)
Defender Gradient Norm $\|\nabla_\theta U_D\|$
Asymptotic Stationarity (Ep. > 175)
Objective / Loss Value
Training Episodes
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0
50
100
150
200
250
(b) Path Neutralization vs. Disruption Budget B_{ops}
Greedy Capacity-Disruption [Alg. 1] (Measured)
Degree-Centrality Revocation (Measured)
Uniform Random Revocation (Measured)
Stackelberg Co-Adaptive Policy (Measured)
Severed Forbidden Paths (%)
Operational Disruption Budget B_{ops} (Revoked Edges)
0
20
40
60
80
100
0
5
10
15
20
25